ARJUNA JEE 2027

Lecture -07 | Mathematics

BINOMIAL THEOREM



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Topics *to be covered*

A **Binomial Series**

B **Multinomial Theorem**

C **Question Practice**

D



H.W. Discussion

Question [JEE Mains-2021]

Let ${}^nC_r$ denote the binomial coefficient of x^r in the expansion of $(1 + x)^n$.

If $\sum_{k=0}^{10}(2^k + 3k)\,{}^{10}C_k = \alpha \cdot 3^{10} + \beta \cdot 2^{10}$, $\alpha, \beta \in R$, $\alpha + \beta$ is equal to _____.

16

85%

$\sum_{r=0}^{10} {}^{10}C_r\, 2^r + 3\sum_{r=0}^{10} r \cdot {}^{10}C_r$

$r \cdot {}^{10}C_r = 10 \cdot {}^{9}C_{r-1}$

$(1+2)^{10} + 30\sum_{r=1}^{10} {}^{9}C_{r-1}$

$\sum_{r=1}^{10} {}^{9}C_{r-1} = 2^9$

$= \underset{\alpha}{1} \cdot 3^{10} + \underset{\beta}{15} \times 2^{10}$

Question [JEE Mains-2023]

If $\frac{1}{n+1}\,{}^{n}C_{n}+\frac{1}{n}\,{}^{n}C_{n-1}+\ldots+\frac{1}{2}\,{}^{n}C_{1}+\underline{{}^{n}C_{0}}=\frac{1023}{10}$, then n is equal to

Ans: A

A 9

B 6

C 7

D 8

$$\sum_{r=0}^{n}\frac{{}^{n}C_{r}}{r+1}$$ ✓

Done!!

Acidic Question

$$S = 3 \cdot \frac{{}^{10}C_0}{1} + 3^2 \cdot \frac{{}^{10}C_1}{2} + 3^3 \cdot \frac{{}^{10}C_2}{3} + \ldots + 3^{11} \cdot \frac{{}^{10}C_{10}}{11}$$

$$S = \sum_{r=0}^{10} \frac{{}^{10}C_r}{r+1} 3^{r+1}$$

$$\frac{1}{11} \sum_{r=0}^{10} {}^{11}C_{r+1} 3^{r+1} = \frac{1}{11}\left\{(1+3)^{11} - 1\right\} = \frac{4^{11}-1}{11} \text{ Ans.}$$

$${}^{11}C_0 3^0 + {}^{11}C_1 3^1 + {}^{11}C_2 3^2 + {}^{11}C_3 3^3 + \ldots + {}^{11}C_{11} 3^{11} - {}^{11}C_0 3^0$$

Question [IIT-JEE-2010]

$A_r = {}^{10}C_r \quad B_r = {}^{20}C_r \quad C_r = {}^{30}C_r$.

For $r = 0, 1, \ldots, 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions $(1 + x)^{10}, (1 + x)^{20}$ and $(1 + x)^{30}$.

60%

Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to

$\sum_{r=1}^{10} A_r B_{10} B_r - \sum_{r=1}^{10} A_r C_{10} A_r$

$= {}^{20}C_{10} \sum_{r=1}^{10} {}^{10}C_r \, {}^{20}C_r - {}^{30}C_{10} \sum_{r=1}^{10} {}^{10}C_r \, {}^{10}C_r$

A $B_{10} - C_{10}$

B $A_{10}(B_{10}^2 - C_{10}A_{10})$

C 0

D $C_{10} - B_{10}$

$${}^{20}C_{10}\left[\sum_{r=1}^{10} {}^{10}C_r\ {}^{20}C_r\right] - {}^{30}C_{10}\left[\sum_{r=1}^{10} {}^{10}C_r\ {}^{10}C_r\right]$$

$$\sum_{r=1}^{10} {}^{10}C_r\ {}^{10}C_{10-r}$$

$$\left({}^{10}C_{10}\,{}^{20}C_0 + {}^{10}C_9\,{}^{20}C_1 + {}^{10}C_8\,{}^{20}C_2 + \ldots + {}^{10}C_0\,{}^{20}C_{10} - {}^{10}C_{10}\,{}^{20}C_0\right)$$

$${}^{10}C_0\,{}^{10}C_{10} + {}^{10}C_1\,{}^{10}C_9 + {}^{10}C_2\,{}^{10}C_8 + \ldots + {}^{10}C_{10}\,{}^{10}C_0 - {}^{10}C_0\,{}^{10}C_{10}.$$

$${}^{30}C_{10} - 1$$

$${}^{20}C_{10}\left({}^{30}C_{10} - 1\right) - {}^{30}C_{10}\left({}^{20}C_{10} - 1\right)$$

$${}^{20}C_{10}\,{}^{30}C_{10} - {}^{20}C_{10} - {}^{30}C_{10}\,{}^{20}C_{10} + {}^{30}C_{10} = {}^{30}C_{10} - {}^{20}C_{10}$$

$$= C_{10} - B_{10}.$$

Type-05:: Replacement

$ax^2 + bx + a \Rightarrow x \to \frac{1}{x}$

$ax^2 + bx + c \Rightarrow x \to \frac{c}{ax} \equiv \frac{\text{Product of Roots}}{x}$

Hint

$(\text{Quad.})^n$

If n is a positive integer and $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$

Then show that :

(i) $a_r = a_{2n-r}$; $0 \le r \le 2n$

(ii) $a_0 + a_1 + a_2 + ... + a_{2n} = 3^n$

(iii) $a_0 - a_1 + a_2 - a_3 + ... + a_{2n} = 1$

(iv) $a_0 + a_1 + a_2 + ... + a_{n-1} = \frac{1}{2}(3^n - a_n)$

(v) $a_0 + a_2 + a_4 + ... + a_{2n} = \frac{1}{2}(3^n + 1)$

(vi) [IIT-JEE 1994] $a_0^2 - a_1^2 + a_2^2 - a_3^2 + ... + a_{2n}^2 = a_n$

$(1+x+x^2)^n = a_0 + a_1x^1 + a_2x^2 + ... + a_{2n}x^{2n}$

$x=1$, $x=-1$

$3^n = a_0 + a_1 + a_2 + ... + a_{n-1} + a_n + a_{n+1} + ... + a_{2n-2} + a_{2n-1} + a_{2n}$

$a_{n-1} + a_n + a_{n+1}$

HHP.

$3^n = 2(a_0 + a_1 + a_2 + ... + a_{n-1}) + a_n$

$a_0a_{2n} - a_1a_{2n-1} + a_2a_{2n-2} - a_3a_{2n-3} .... a_{2n}a_0$

$$(1+x+x^2)^n = a_0 + a_1x^1 + a_2x^2 + \dots\dots + a_{2n}x^{2n} \quad \text{— A}$$

$x \to \frac{1}{x}$

$$\frac{(x^2+x+1)^n}{x^{2n}} = \frac{a_0x^{2n} + a_1x^{2n-1} + a_2x^{2n-2} + a_3x^{2n-3} + \dots\dots + a_{2n}}{x^{2n}}$$

same

$$(x^2+x+1)^n = a_0x^{2n} + a_1x^{2n-1} + a_2x^{2n-2} + a_3x^{2n-3} + \dots + a_{2n-1}x^1 + a_{2n} \quad \text{— B}$$

$a_0 = a_{2n}$ $\quad a_1 = a_{2n-1}$ — — $\Rightarrow$ $\boxed{a_r = a_{2n-r}}$

$$\left(1+x+x^2\right)^n = a_0 + a_1x^1 + a_2x^2 + \dots\dots + a_{2n}x^{2n}$$

$x \to -x$

$$\left(1-x+x^2\right)^n = a_0 - a_1x^1 + a_2x^2 - \dots\dots + a_{2n}x^{2n}$$

$$\left(1+x+x^2\right)^n\left(1-x+x^2\right)^n = \left(a_0 + a_1x^1 + a_2x^2 + \dots\dots + a_{2n}x^{2n}\right)\left(a_0 - a_1x^1 + a_2x^2 - \dots - a_{2n-1}x^{2n-1} + a_{2n}x^{2n}\right)$$

coeff. of x^{2n} $\left(a_0a_{2n} - a_1a_{2n-1} + a_2a_{2n-2} - \dots\dots\right)x^{2n}$

coeff. of x^{2n} in $\left(1+x+x^2\right)^n\left(1-x+x^2\right)^n$

$$\left(1+x+x^2\right)^n = a_0 + a_1x^1 + a_2x^2 + \dots\dots + a_{2n}x^{2n}$$

coeff. of x^{2n} in $\left(1+x+x^2\right)^n\left(1-x+x^2\right)^n$

coeff. of x^{2n} in $\left(1+x^2+x^4\right)^n = a_n$

$\left(1+x+x^2\right)\left(1-x+x^2\right) \subseteq 1+x^2+x^4.$

$$\left(1+x+x^2\right)^n = a_0 + a_1x^1 + a_2x^2 + \dots + a_nx^n + \dots\dots + a_{2n}x^{2n}$$

$x \to x^2$

$$\left(1+x^2+x^4\right)^n = a_0 + a_1x^2 + a_2x^4 + \dots + a_nx^{2n} + \dots\dots + a_{2n}x^{4n}$$

$$\left(1+x+x^2\right)^n = a_0 + a_1x^1 + a_2x^2 + a_3x^3 \ldots + a_{2n}x^{2n}$$

"diff. both side"

① $a_1 + 2a_2 + 3a_3 + \ldots + 2n\,a_{2n} = n \cdot 3^n$

② $a_1 - 2a_2 + 3a_3 - \ldots - 2n\,a_{2n}$

$$n\left(1+x+x^2\right)^{n-1} \times (1+2x) = 1a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \ldots + 2n\,a_{2n}x^{2n-1}$$

$x = 1$

$$n \times 3^{n-1} \times 3^1 = a_1 + 2a_2 + 3a_3 + \ldots + 2n\,a_{2n}$$

$x = -1$

$$-n = a_1 - 2a_2 + 3a_3 - \ldots - 2n\,a_{2n}$$

DOUBLE SIGMA PROBLEMS

Important Formula : # If $f(i,j)$ → Symmetric func^n of i & j

$$\sum_{j=0}^{n}\sum_{i=0}^{n} f(i,j) = \sum_{i<j}\sum f(i,j) + \sum_{j<i}\sum f(i,j) + \sum_{i=j}\sum f(i,j)$$

Total — equal — $j \to i$

$$\sum_{i=m}^{\sim} f(i,i)$$

- $i^2 + j^2$ ✓
- ${}^nC_i + {}^nC_j$ ✓
- ${}^nC_i \, {}^nC_j$ ✓

$$\frac{\text{Total} - (i=j)}{2} = i<j = j<i$$

IMPORTANT NOTE

Dhyaan se Dekho

$$\sum_{j=0}^{3}\sum_{i=0}^{3}(i^2+j^2)=$$

(0^2+0^2) (1^2+0^2) (2^2+0^2) (3^2+0^2)

(0^2+1^2) (1^2+1^2) (2^2+1^2) (3^2+1^2)

(0^2+2^2) (1^2+2^2) (2^2+2^2) (3^2+2^2)

(0^2+3^2) (1^2+3^2) (2^2+3^2) (3^2+3^2)

Bumper Question

Find all of the following:

$$P=\sum_{j=0}^{n}\sum_{i=0}^{n}({}^{n}C_i+{}^{n}C_j)$$

$$=\sum_{j=0}^{n}\left(\sum_{i=0}^{n}{}^{n}C_i+\sum_{i=0}^{n}{}^{n}C_j\right)$$

($\sum_{i=0}^{n}{}^{n}C_i = 2^n$, $\sum_{i=0}^{n}{}^{n}C_j = (n+1)\,{}^{n}C_j$)

$$=\sum_{j=0}^{n}\left(2^n+(n+1)\cdot{}^{n}C_j\right)$$

$$Q=\sum_{j=0}^{n}\sum_{i=0}^{n}({}^{n}C_i+{}^{n}C_j)$$ (where $i=j$)

$j \to i$

$$\sum_{i=0}^{n}\left({}^{n}C_i+{}^{n}C_i\right)=2\sum_{i=0}^{n}{}^{n}C_i \quad (\sum_{i=0}^{n}{}^{n}C_i = 2^n)$$

$$=2\cdot 2^n=\boxed{2^{n+1}=Q}$$

$$\sum_{j=0}^{n}2^n+(n+1)\sum_{j=0}^{n}{}^{n}C_j \quad (\sum_{j=0}^{n}{}^{n}C_j = 2^n)$$

$$(n+1)2^n+(n+1)2^n$$

$$2(n+1)2^n=\boxed{(n+1)\cdot 2^{n+1}=P}$$

$$R=\sum_{0\le i<j\le n}\sum({}^{n}C_i+{}^{n}C_j)$$

$$S=\sum_{0\le i\le j\le n}\sum({}^{n}C_i+{}^{n}C_j)$$

Bumper Question

$$P = Q + R + R$$

Find all of the following:

$$P = \sum_{j=0}^{n}\sum_{i=0}^{n}({}^nC_i + {}^nC_j)$$

$$Q = \sum_{j=0}^{n}\sum_{i=0}^{n}({}^nC_i + {}^nC_j) \quad (\text{where } i=j)$$

$$R = \sum\sum_{0\le i< j\le n}({}^nC_i + {}^nC_j)$$

$$S = \sum\sum_{0\le i\le j\le n}({}^nC_i + {}^nC_j)$$

$i \le j \Rightarrow (i<j) + (i=j)$ — $(i<j)$: R, $(i=j)$: Q

$$R_{(i<j)} = \frac{\text{Total} - (i=j)}{2} = \frac{P-Q}{2}$$

$$R = \frac{(n+1)2^{n+1} - 2^{n+1}}{2} = \frac{2^{n+1}(n)}{2} = n2^n$$

$$\#\xi = R + Q$$

$$\#(i>j) = R$$

$$\#(i\neq j) = P - Q = 2R$$

$(i\neq j) = (i>j) + (i<j)$

Question [JEE Mains-2022]

$\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^{n}C_i \, {}^{n}C_j$ is equal to

(A) $2^{2n} - {}^{2n}C_n$

(B) $2^{2n-1} - {}^{2n-1}C_{n-1}$

(C) $2^n - \frac{1}{2}{}^{2n}C_n$

(D) $2^{n-1} + {}^{2n-1}C_n$

$$\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^{n}C_i \, {}^{n}C_j = \sum_{i=0}^{n}\sum_{\substack{j=0 \\ (i\neq j)}}^{n} {}^{n}C_i \, {}^{n}C_j$$

dew.

MULTINOMIAL THEOREM

TRI-NOMIAL :

$$(a+b+c)^n = \sum_{\alpha+\beta+\gamma=n} \frac{n!}{\alpha!\,\beta!\,\gamma!}\, a^\alpha b^\beta c^\gamma \quad , \quad \alpha, \beta, \gamma \in W$$

$\hookrightarrow$ n.o.t $= {}^{n+3-1}C_{3-1}$ * (soch smjh kr use)

$$(a+b+c+d)^n = \sum_{\alpha+\beta+\gamma+\delta=n} \frac{n!}{\alpha!\,\beta!\,\gamma!\,\delta!}\, a^\alpha b^\beta c^\gamma d^\delta \quad , \quad \alpha, \beta, \gamma, \delta \in W$$

$\hookrightarrow$ n.o.t $= {}^{n+4-1}C_{4-1}$ * (soch smjh kr use)

$(x+y+z)^{10}$

$\hookrightarrow$ n.o.t. $= {}^{10+3-1}C_{3-1} = {}^{12}C_2$ ✓

$(x^2+x+1)^{10}$

$\hookrightarrow$ n.o.t. $= {}^{10+3-1}C_{3-1} = {}^{12}C_2$ ✗

Question

The coefficient of x^4 in $(3x^2 - x - 2)^{10}$ is _____.

M-1

$(x-1)^{10}(3x+2)^{10}$

x^0	x^4
x^1	x^3
x^2	x^2
x^3	x^1
x^4	x^0

$$G.T. = \frac{10!}{\alpha!\,\beta!\,\gamma!}(3x^2)^{\alpha}(-x)^{\beta}(-2)^{\gamma}$$

$$G.T. = \frac{10!}{\alpha!\,\beta!\,\gamma!}3^{\alpha}(-1)^{\beta}(-2)^{\gamma}x^{2\alpha+\beta}$$

$$\frac{10!}{2!\,0!\,8!}3^2(-1)^0(-2)^8 + \frac{10!}{1!\,2!\,7!}3^1(-1)^2(-2)^7 + \frac{10!}{0!\,4!\,6!}3^0(-1)^4(-2)^6$$

$\alpha + \beta + \gamma = 10$.

$2\alpha + \beta = 4$ & $\alpha + \beta + \gamma = 10$.

α	β	γ
2	0	8
1	2	7
0	4	6

Question [JEE Mains-2023]

The coefficient of x^7 in $(1-x+2x^3)^{10}$ is

Ans: 960

n.o.t. $= {}^{10+3-1}C_{3-1} = {}^{12}C_2$ ✗

$\beta+3\gamma=7$ & $\alpha+\beta+\gamma=10$

α	β	γ
3	7	0
5	4	1
7	1	2

$$\frac{10!}{\alpha!\beta!\gamma!}(1)^{\alpha}(-x)^{\beta}(2x^3)^{\gamma}$$

$$\frac{10!}{\alpha!\beta!\gamma!}(-1)^{\beta}\cdot 2^{\gamma}\cdot x^{\beta+3\gamma}$$

$$960 = \frac{10!}{3!7!0!}(-1)^7 2^0 + \frac{10!}{5!4!1!}(-1)^4\cdot 2^1 + \frac{10!}{7!1!2!}(-1)^1\cdot 2^2$$

Question [JEE Mains-2023]

The constant term in the expansion of $\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is

Ans: 1080

$$\frac{5!}{\alpha!\,\beta!\,\gamma!}\,(2x)^{\alpha}\cdot(x^{-7})^{\beta}(3x^2)^{\gamma}$$

$$\frac{5!}{\alpha!\,\beta!\,\gamma!}\,2^{\alpha}\,3^{\gamma}\,x^{\alpha-7\beta+2\gamma}$$

$\alpha-7\beta+2\gamma = 0$ & $\alpha+\beta+\gamma = 5$

α	β	γ
1	1	3

$$\frac{5!}{1!\,1!\,3!}\,2^1\,3^3 = 5\cdot 4\cdot 2\cdot 27$$

$$= 1080 \checkmark$$

Question [JEE Mains-2021]

If the coefficient of a^7b^8 in the expansion of $(a + 2b + 4ab)^{10}$ is $K.2^{16}$, then K is equal to_____.



Question [JEE Mains-2020]

Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to _____.

a_7 → Coeff of x^7

a_{13} → Coeff of x^{13}



Replacement

$x \to \frac{a}{x}$

Trinomial

DO IT BY YOURSELF (DIBY)

FINDING COEFFICIENTS

DIBY-43

The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^6$ in powers of x, is

[JEE Mains-2020]

[Ans. : (120)]

DIBY-44

The term independent of x in the expansion of $(1-x^2+3x^3)\left(\frac{5}{2}x^3-\frac{1}{5x^2}\right)^{11}$, $x \neq 0$ is:

[JEE Mains-2022]

(A) $\frac{7}{40}$ (B) $\frac{33}{200}$ (C) $\frac{39}{200}$ (D) $\frac{11}{50}$

[Ans. : (B)]

DIBY-45

If $\left({}^{40}C_0\right)+\left({}^{41}C_1\right)+\left({}^{42}C_2\right)+\ldots.+\left({}^{60}C_{20}\right)=\frac{m}{n}{}^{60}C_{20}$, m and n re coprime, then $m+n$ is equal to

[JEE Mains-2022]

[Ans. : 102]

DIBY-46

If $1^2\cdot\left({}^{15}C_1\right)+2^2\cdot\left({}^{15}C_2\right)+3^2\cdot\left({}^{15}C_3\right)+\ldots+15^2\cdot\left({}^{15}C_{15}\right)=2^m.3^n.5^k$, where $m,n,k\in N$, then $m+n+k$ is equal to

(A) 19 (B) 21 (C) 18 (D) 20

[Ans. : (A)]

DIBY-47

The sum of series $2\times1\times{}^{20}C_4-3\times2\times{}^{20}C_5+4\times3\times{}^{20}C_6-5\times4\times{}^{20}C_7+\ldots+18\times17\times{}^{20}C_{20}$ is equal to _____

[JEE Mains-2025]

[Ans. : 37]

DIBY-48

$\sum_{r=0}^{5} \frac{^{11}C_{2r+1}}{2r+2} = \frac{m}{n}, \gcd(m,n) = 1$, then $m - n$ is equal to [JEE Mains-2025]

[Ans. : 2035]

DIBY-49

If $\sum_{r=1}^{30} \frac{r^2\left(^{30}C_r\right)^2}{^{30}C_{r-1}} = \alpha \times 2^{29}$, then α is equal to [JEE Mains-2025]

[Ans. : 465]

DIBY-50

Let $\alpha = \sum_{r=0}^{n} \left(4r^2 + 2r + 1\right)^n C_r$ and $\beta = \left(\sum_{r=0}^{n} \frac{^nC_r}{r+1}\right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is [JEE Mains-2024]

[Ans. : 5

SERIES

DIBY-51

The value of $\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+\ldots+\frac{1}{49!22!}+\frac{1}{51!1!}$ is [JEE Mains-2023]

(A) $\frac{2^{50}}{50!}$ (B) $\frac{2^{50}}{51!}$ (C) $\frac{2^{51}}{51!}$ (D) $\frac{2^{51}}{50!}$

[Ans. : (B)]

DIBY-52

If $\sum_{r=0}^{10}\left(\frac{10^{r+1}-1}{10^r}\right)\cdot {}^{11}C_{r+1}=\frac{\alpha^{11}-11^{11}}{10^{10}}$, then α is equal to: [JEE Mains-2025]

(A) 15 (B) 11 (C) 24 (D) 20

[Ans. : (A)]

DIBY-53

If $\sum_{r=1}^{9}\left(\frac{r+3}{2^r}\right)\cdot {}^{9}C_r=\alpha\left(\frac{3}{2}\right)^9-\beta, \alpha, \beta \in N$, then $(\alpha+\beta)^2$ is equal to [JEE Mains-2025]

(A) 27 (B) 9 (C) 81 (D) 18

[Ans. : (C)]

DIBY-54

If $^{(^{30}C_1)^2} + 2^{(^{30}C_1)^2} + 3^{(^{30}C_3)^2} + \cdots .. + 30^{(^{30}C_1)^2} = \frac{\alpha \cdot 60!}{(30!)^2}$, then α is equal to

[JEE Mains-2023]

(A) 30 (B) 60 (C) 15 (D) 10

[Ans. : (C)]

DIBY-55

If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3 \left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to

[JEE Mains-2023]

(A) 4895 (B) 1210 (C) 5445 (D) 3025

[Ans. : (B)]

DIBY-56

If $1 + \left(2 + {}^{49}C_1 + {}^{49}C_2 + \cdots + {}^{49}C_{49}\right)\left({}^{50}C_2 + {}^{50}C_4 + \cdots {}^{50}C_{50}\right)$ is equal to $2^n \cdot m$, where m is odd, then $n + m$ is equal to ___

[JEE Mains-2022]

[Ans. : 99]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians